A **schema** is the way data is **organized into tables and relationships** in a database or tools like **Microsoft Power BI**.

There are mainly **three types of schemas**:

1. Star Schema

In **Star Schema**, there is:

- One **Fact Table** (main data like Sales, Revenue)
- Multiple **Dimension Tables** (like Customer, Product, Date)
- It looks like a **star shape**, so it is called star schema.

Example:

- Fact Table → Sales
- Dimension Tables → Customer, Product, Time

Advantages:

- Simple and easy to understand
- Fast performance

Disadvantage:

- Data redundancy (same data repeated)

2. Snowflake Schema

In **Snowflake Schema**, dimension tables are **further divided into sub-tables**.

It looks like a snowflake shape.

Example:

- Product → Category → Subcategory

Advantages:

- Less redundancy
- More organized

Disadvantages:

- Complex
- Slower than star schema

3. Galaxy Schema (Fact Constellation)

In this schema:

- There are **multiple fact tables**
- Shared dimension tables

Example:

- Fact Tables → Sales, Orders
- Shared Dimensions → Customer, Product

Advantages:

- Handles complex systems
- Useful for large organizations

Disadvantages:

- Difficult to design and understand

1. Relationship Rationale (Why we create relationships)

A **relationship** connects two tables in tools like **Microsoft Power BI**.

The **rationale (reason)** for creating relationships is:

- To **combine data from multiple tables**
- To avoid duplication
- To maintain **data consistency**
- To enable correct **analysis and reporting**

Example:

- Customer table (CustomerID)
- Orders table (CustomerID)

We connect them using **CustomerID** so we can know:

- Which customer made which order

2. Types of Relationships (Basic)

- **One-to-Many (1:*)** → Most common
- **Many-to-One (*:1)**
- **Many-to-Many (*:*)**

Usually:

- **Primary Key → Foreign Key**

3. Filter Flow (Very Important Concept)

Filter flow means:
How filtering in one table affects another table

4. Direction of Filter Flow

Single Direction (Default)

Filter flows from:
Dimension Table → Fact Table

Example:

- Customer → Orders

If you select:
Customer = “A”

Then:
Orders will show only orders of Customer A

Both Direction (Bi-directional)

Filter flows both ways:

- Customer ↔ Orders

If you filter Orders:
Customer table also gets filtered

5. Why Filter Flow is Important

- Ensures **correct results in reports**
- Controls **how visuals interact**
- Prevents **wrong calculations**

6. Example (Simple)

Tables:

- Customers (CustomerID)
- Orders (CustomerID)

Relationship:
Customers → Orders

Now:

- If you select CustomerID = 1
- Only orders of that customer will show

This is **filter flow working**